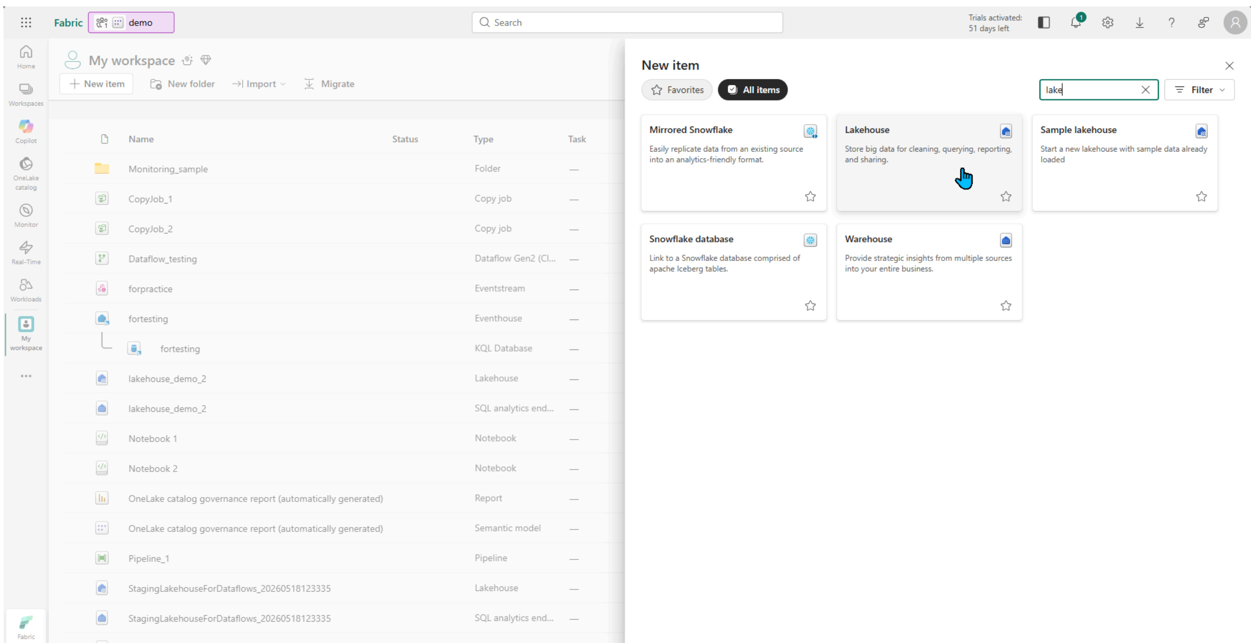
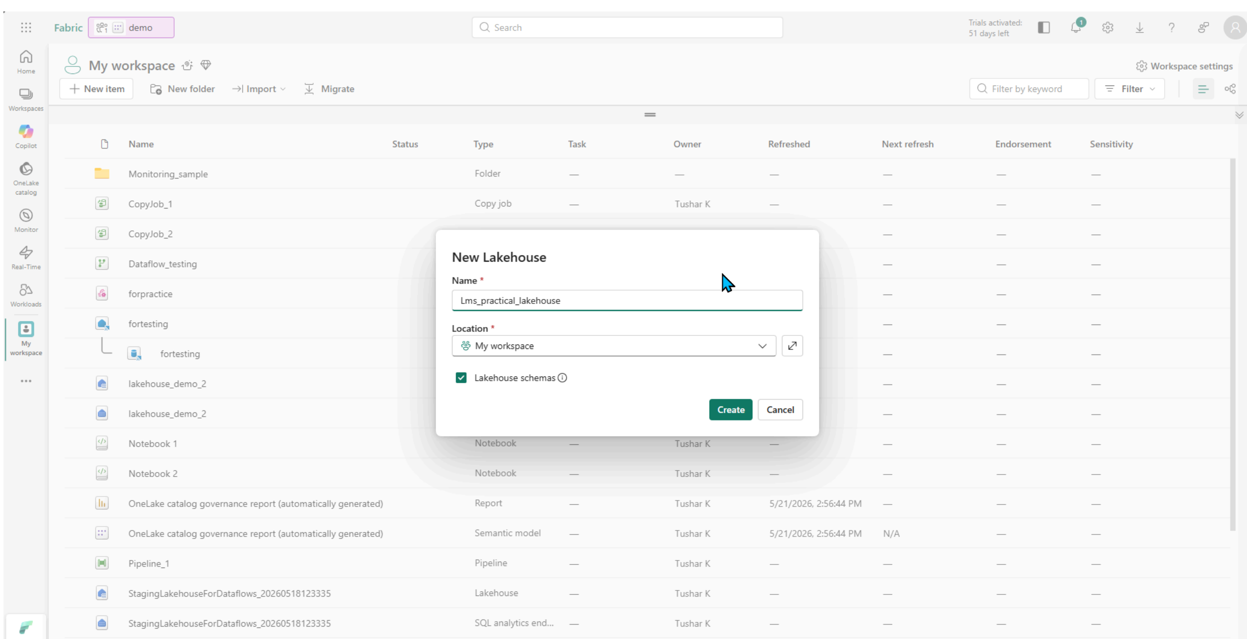
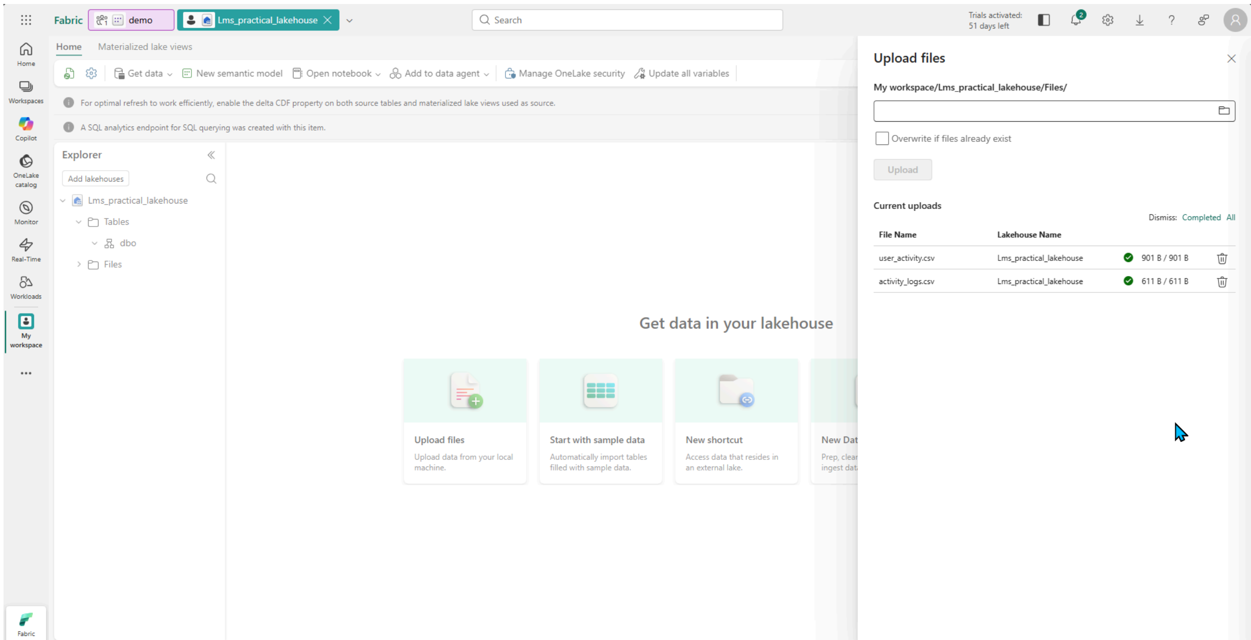
Fabric assignment

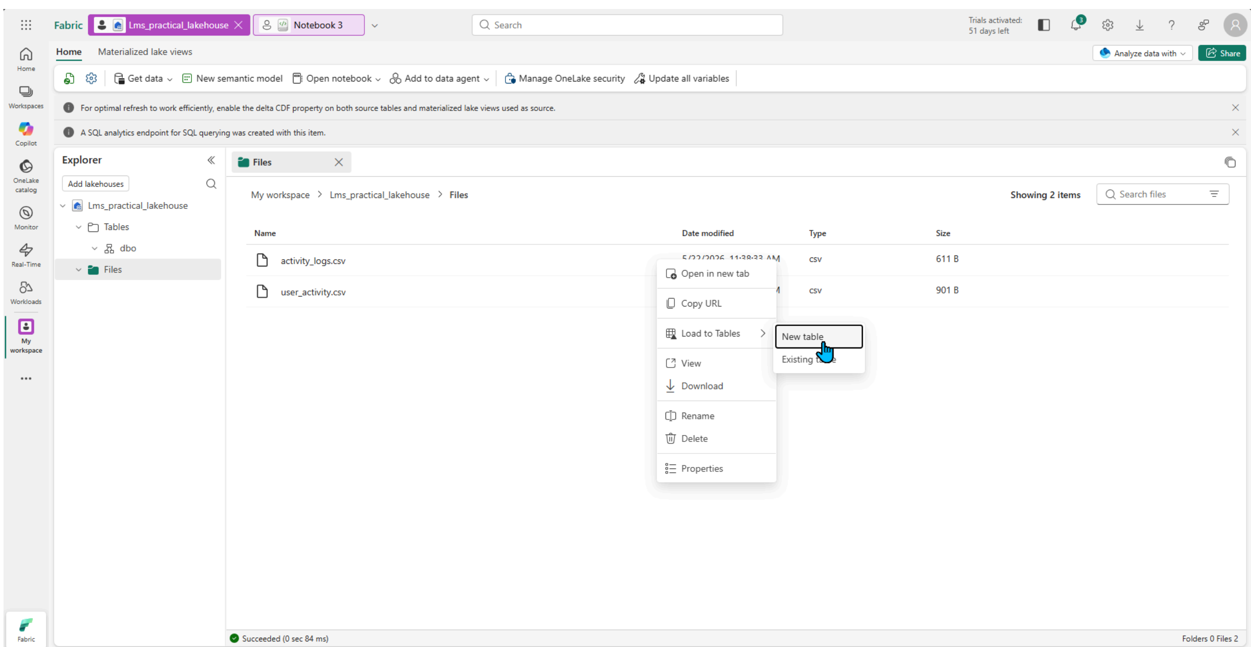
Task 1: OneLake & Lakehouse

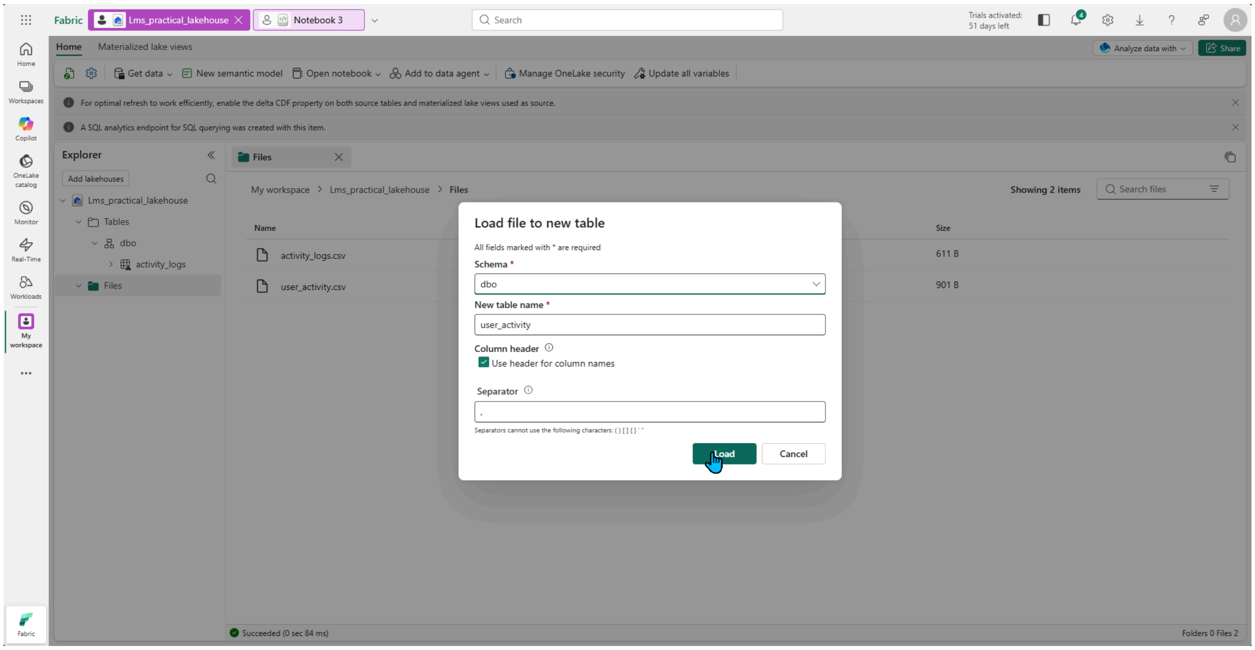
- Create a Lakehouse in Microsoft Fabric
- Upload the provided CSV file to the Files section
- Load the data into Lakehouse Tables
- Verify the table using the SQL Endpoint

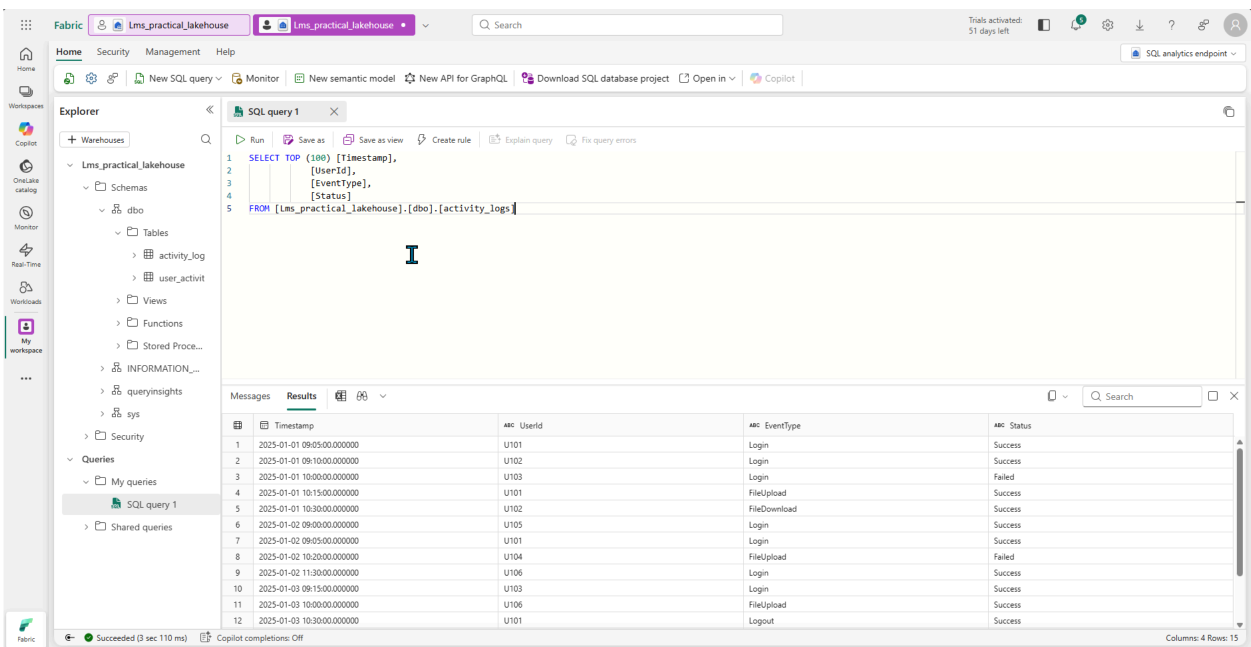


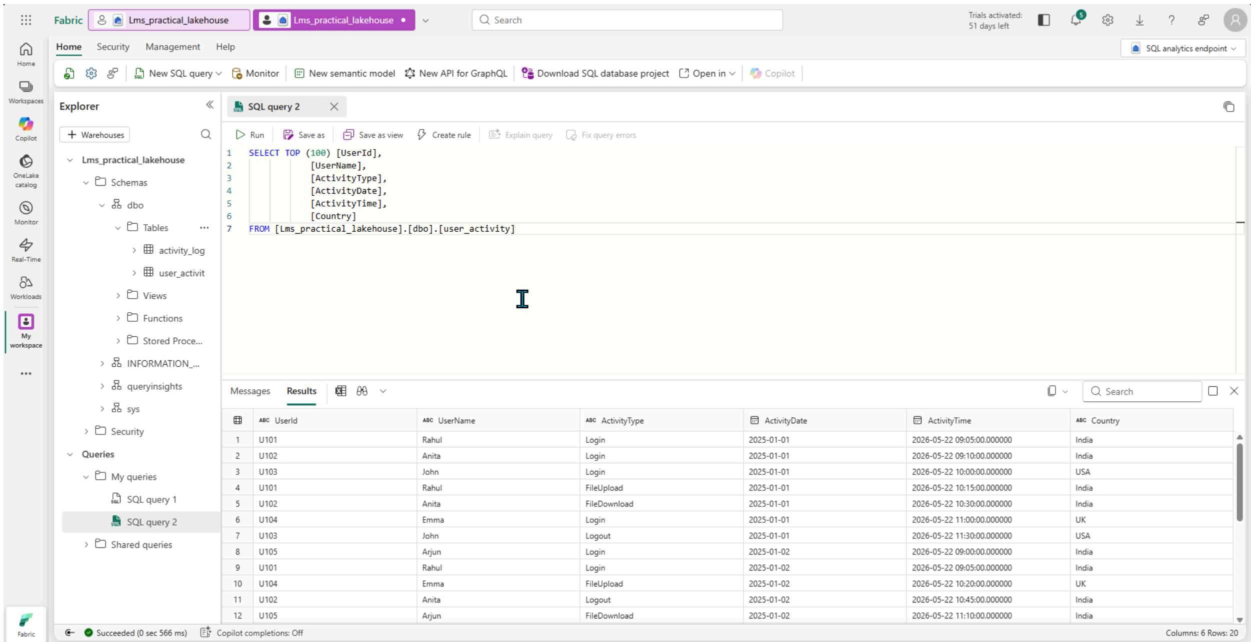






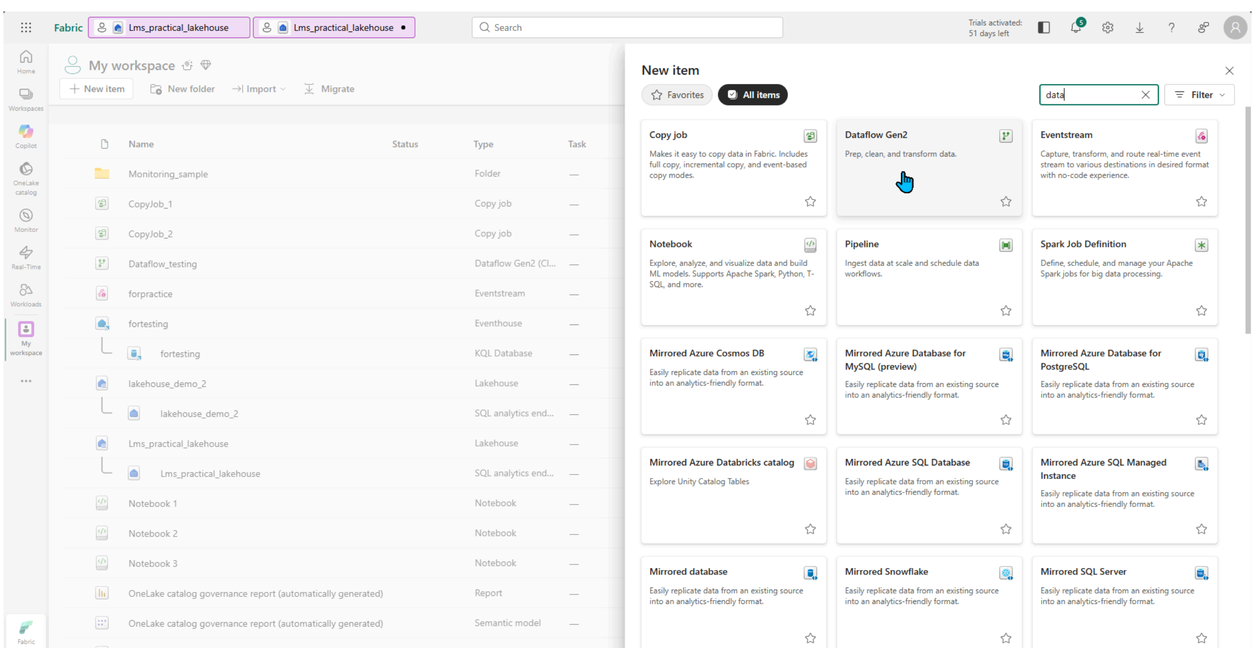

Fabric
Lms_practical_lakehouse
Notebook 3
Search
Trials activated: 51 days left
Home
Materialized lake views
Get data
New semantic model
Open notebook
Add to data agent
Manage OneLake security
Update all variables
Analyze data with
Share
For optimal refresh to work efficiently, enable the delta CDF property on both source tables and materialized lake views used as source.
A SQL analytics endpoint for SQL querying was created with this item.
Explorer
Add lakehouses
Lms_practical_lakehouse
Tables
dbo
activity_logs
Files
My workspace > Lms_practical_lakehouse > Files
Showing 2 items
Search files
Name
Size
activity_logs.csv
611 B
user_activity.csv
901 B
Load file to new table
All fields marked with * are required
Schema *
dbo
New table name *
user_activity
Column header
Use header for column names
Separator
,
Load
Cancel
Succeeded (0 sec 84 ms)
Folders 0 Files 2

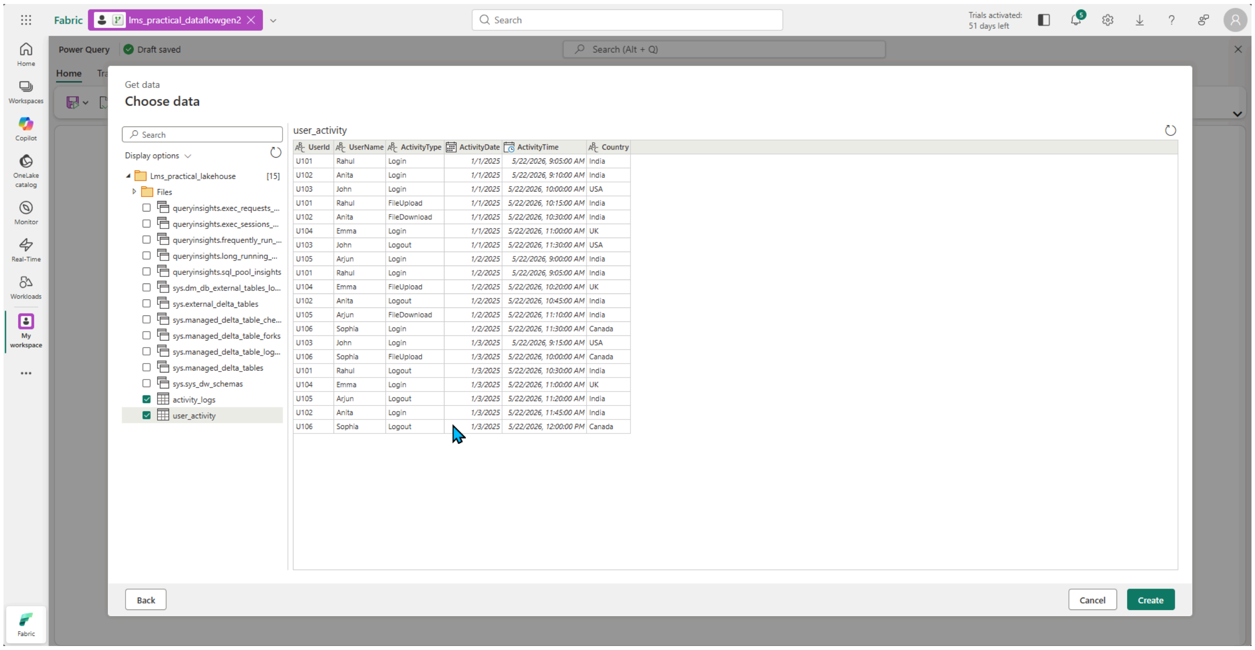

Fabric
Lms_practical_lakehouse
Lms_practical_lakehouse
Search
Trials activated: 51 days left
Home
Security
Management
Help
SQL analytics endpoint
New SQL query
Monitor
New semantic model
New API for GraphQL
Download SQL database project
Open in
Copilot
Explorer
Warehouses
Lms_practical_lakehouse
Schemas
dbo
Tables
activity_log
user_activit
Views
Functions
Stored Proce...
INFORMATION_...
queryinsights
sys
Security
Queries
My queries
SQL query 1
Shared queries
SQL query 1
Run
Save as
Save as view
Create rule
Explain query
Fix query errors
SELECT TOP (100) [Timestamp],
[UserId],
[EventType],
[Status]
FROM [Lms_practical_lakehouse].[dbo].[activity_logs]
Messages
Results
Timestamp
UserId
EventType
Status
2025-01-01 09:05:00.000000 U101 Login Success
2025-01-01 09:10:00.000000 U102 Login Success
2025-01-01 10:00:00.000000 U103 Login Failed
2025-01-01 10:15:00.000000 U101 FileUpload Success
2025-01-01 10:30:00.000000 U102 FileDownload Success
2025-01-02 09:00:00.000000 U105 Login Success
2025-01-02 09:05:00.000000 U101 Login Success
2025-01-02 10:20:00.000000 U104 FileUpload Failed
2025-01-02 11:30:00.000000 U106 Login Success
2025-01-03 09:15:00.000000 U103 Login Success
2025-01-03 10:00:00.000000 U106 FileUpload Success
2025-01-03 10:30:00.000000 U101 Logout Success
Succeeded (3 sec 110 ms)
Copilot completions: Off
Columns: 4 Rows: 15

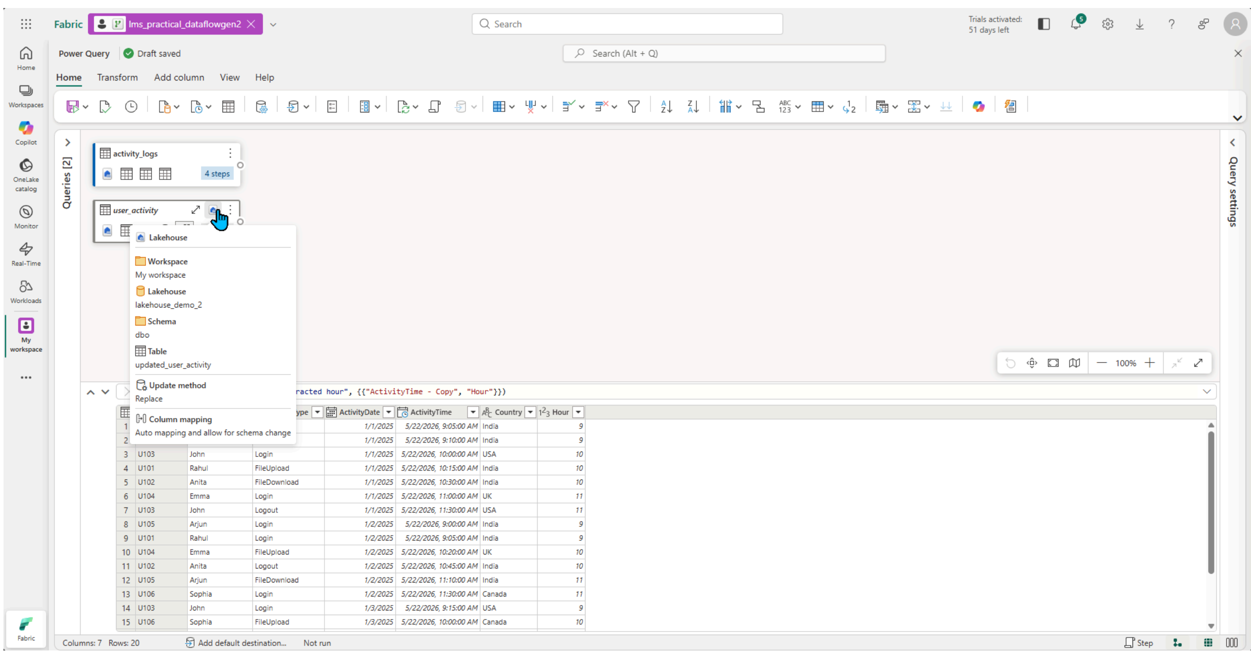


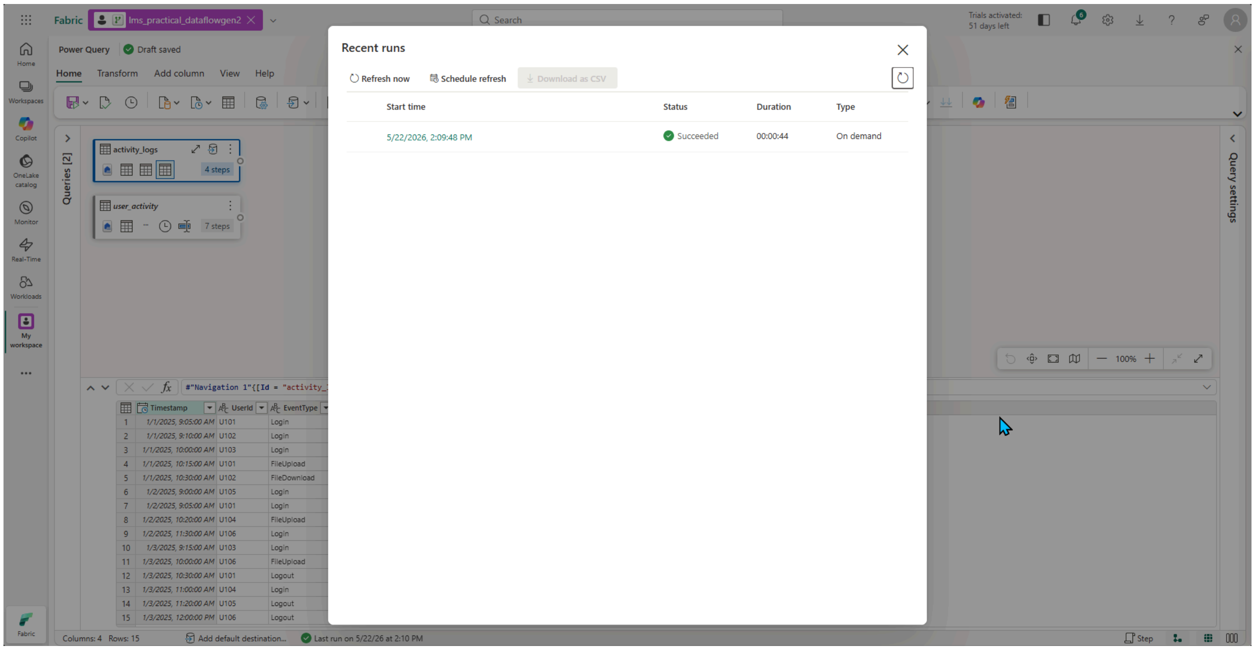
Task 2: Dataflow Gen2

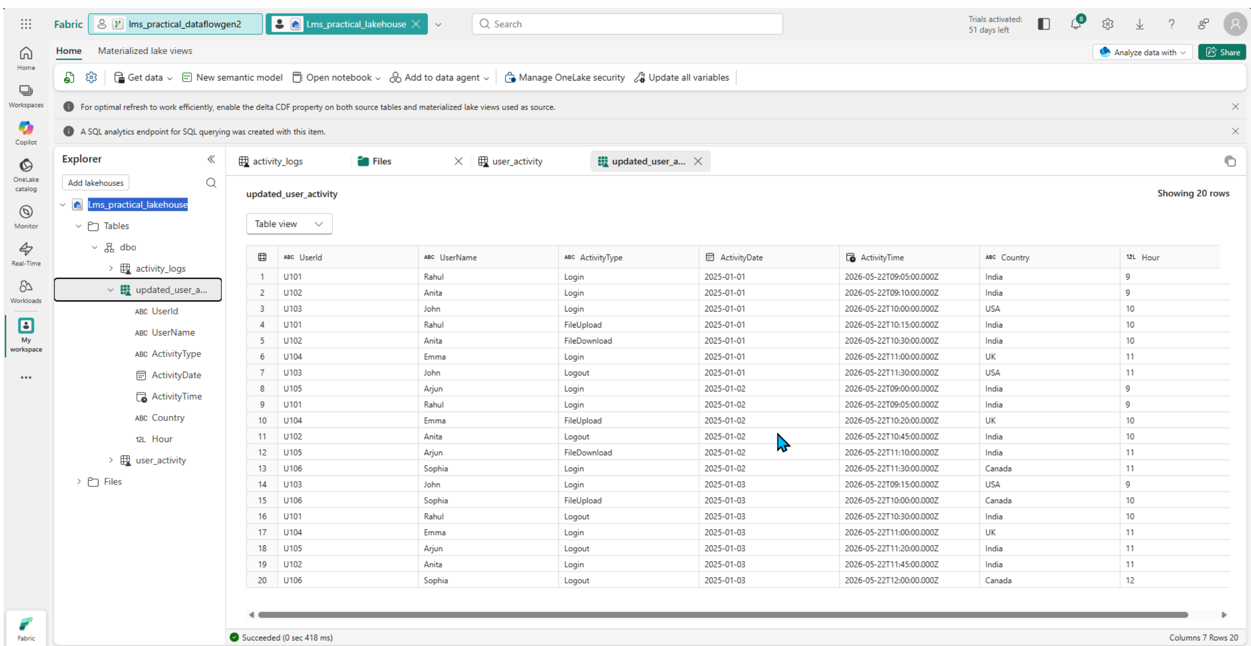
- Create a Dataflow Gen2
- Use Lakehouse table or CSV as the source
- Perform basic transformations:
  a) Convert activity date to Date format
  b) Derive ActivityHour from activity time
  c) Rename columns for clarity
- Load the transformed data back to the same Lakehouse as a new table.




Fabric
lms_practical_dataflowgen2
Power Query
Draft saved
Get data
Choose data
Display options
Lms_practical_lakehouse
Files
activity_logs
user_activity
Back
Cancel
Create

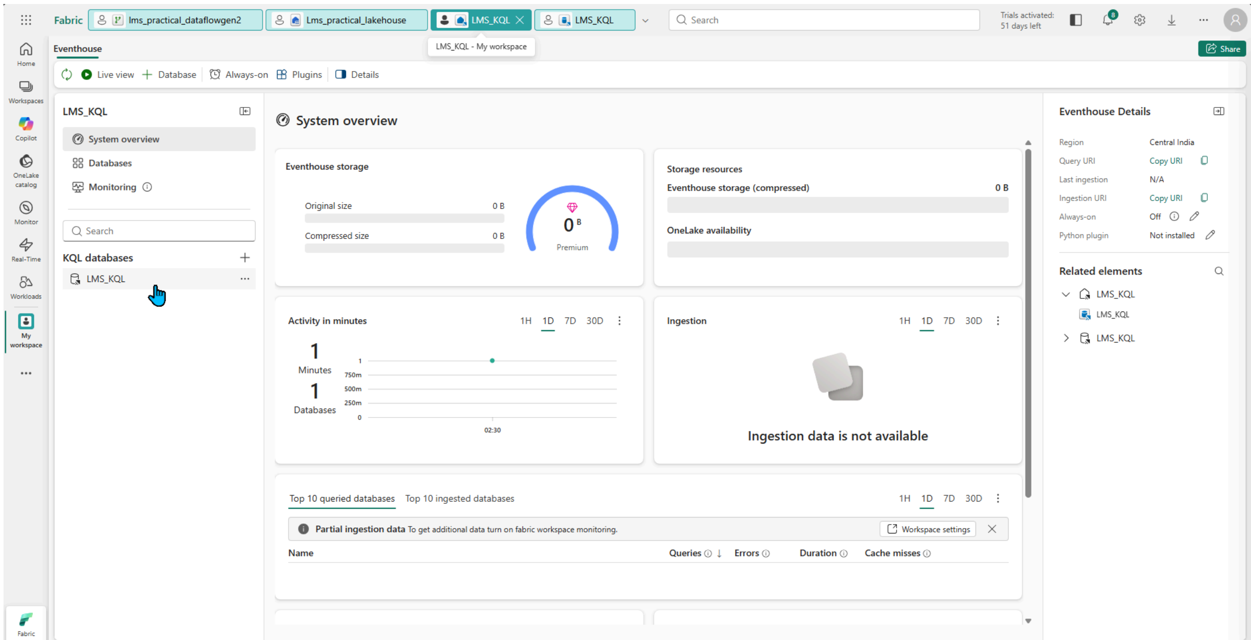

Fabric
lms_practical_dataflowgen2
Power Query
Draft saved
Home
Transform
Add column
View
Help
activity_logs
4 steps
user_activity
Lakehouse
Workspace
My workspace
Lakehouse
lakehouse_demo_2
Schema
dbo
Table
updated_user_activity
Update method
Replace
Column mapping
Auto mapping and allow for schema change
Query settings
Columns: 7 Rows: 20
Add default destination...
Not run

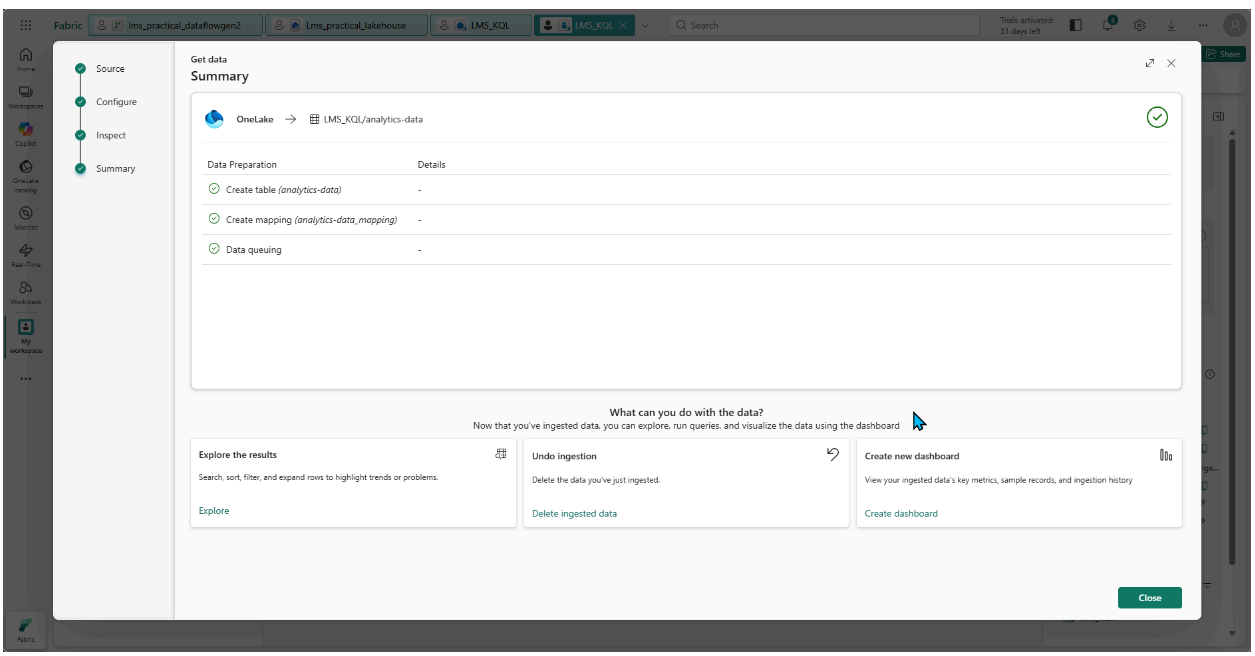


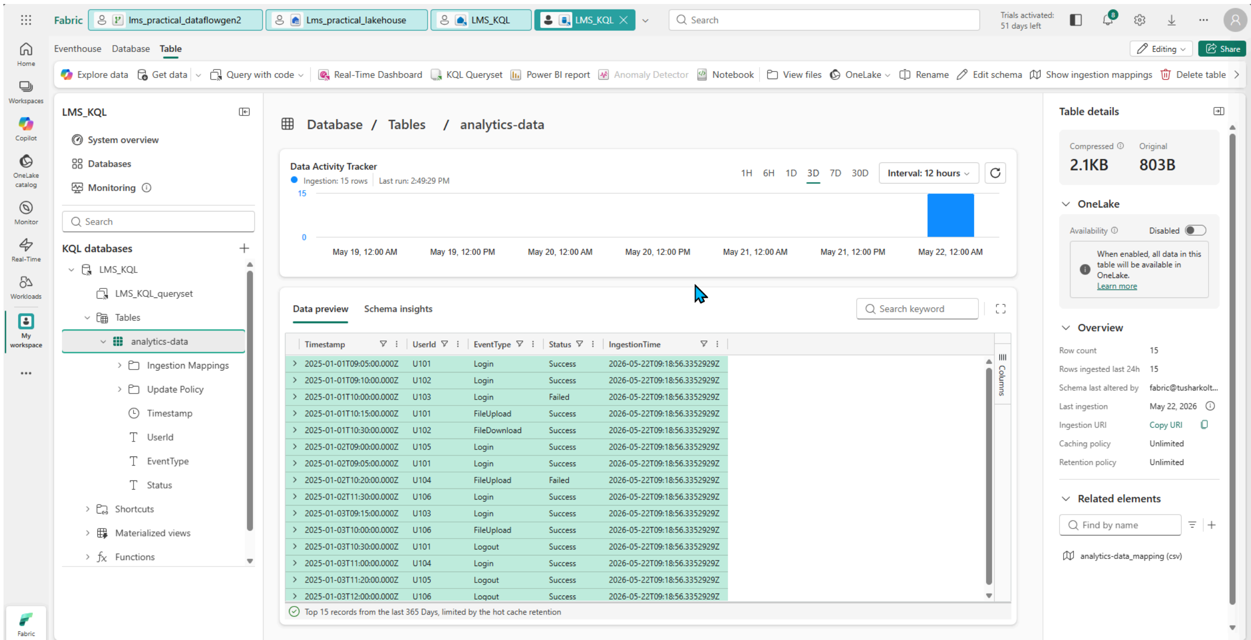


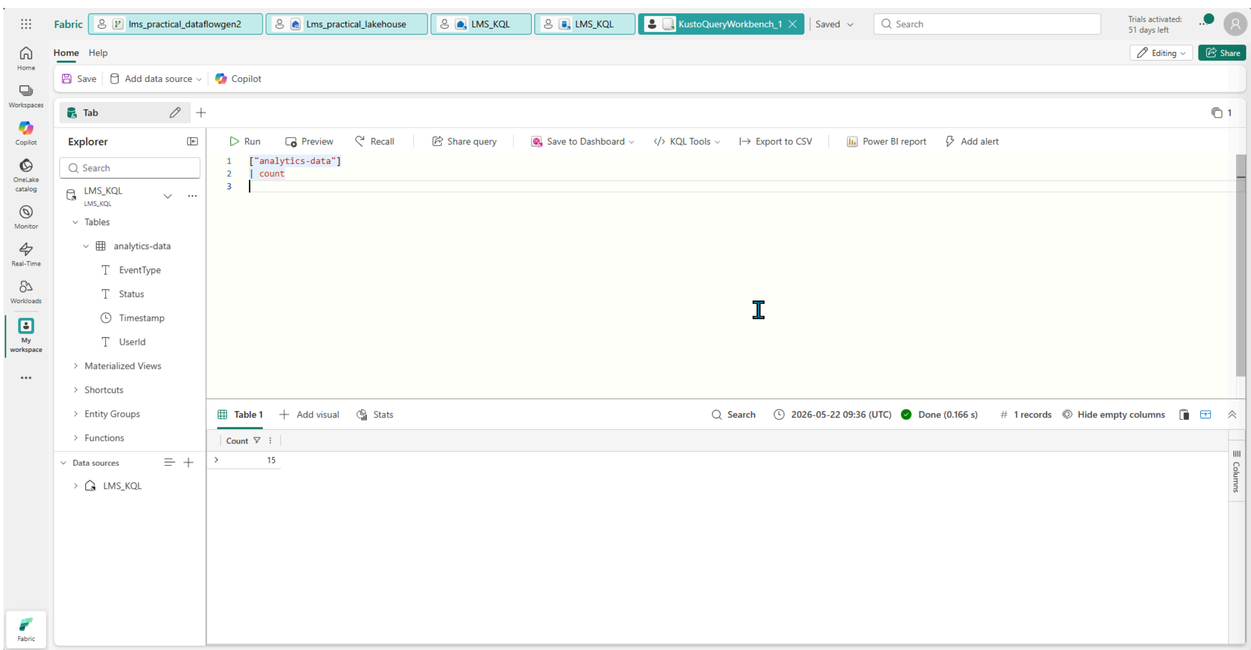
Task 3: KQL Analytics

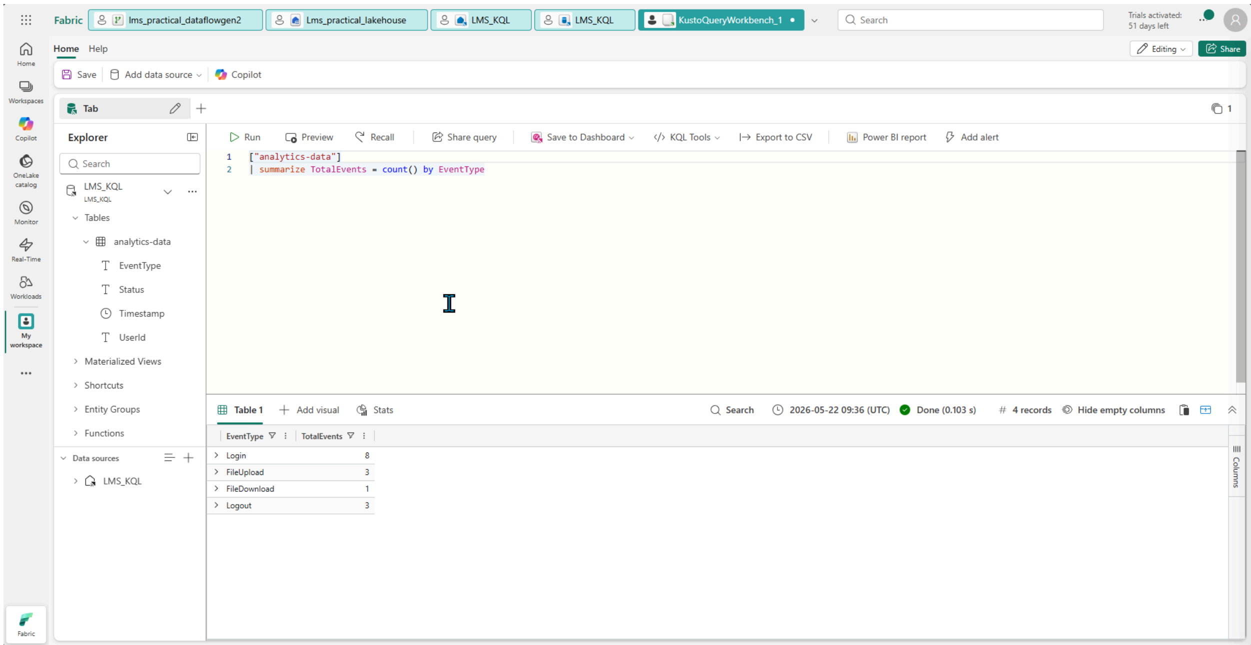
- Create a KQL Database in Fabric
- Load sample or generated log data
- Write KQL queries to analyze:
  a) Total event count
  b) Events by event type
  c) Successful vs failed logins

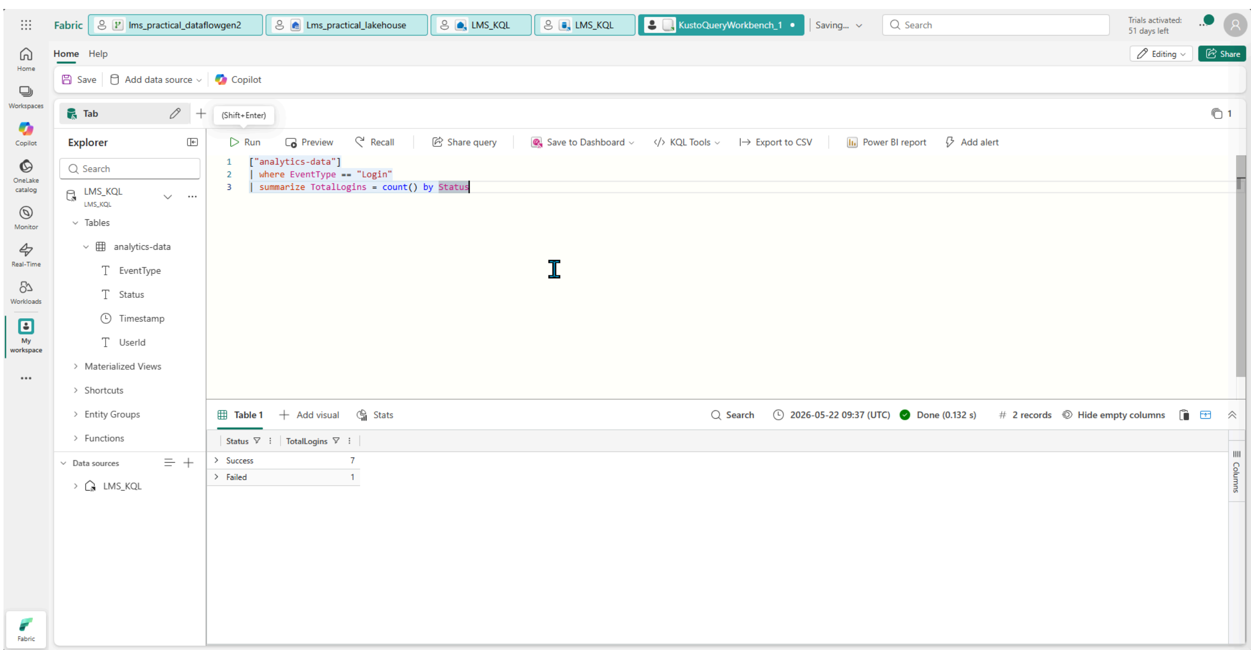

Eventhouse
System overview
Eventhouse storage
Storage resources
Activity in minutes
Ingestion
Ingestion data is not available
Top 10 queried databases
Top 10 ingested databases
Eventhouse Details
Related elements


Get data
Summary
Source
Configure
Inspect
Summary
OneLake
LMS_KQL/analytics-data
Data Preparation
Details
Create table (analytics-data)
Create mapping (analytics-data_mapping)
Data queuing
What can you do with the data?
Now that you've ingested data, you can explore, run queries, and visualize the data using the dashboard
Explore the results
Search, sort, filter, and expand rows to highlight trends or problems.
Explore
Undo ingestion
Delete the data you've just ingested.
Delete ingested data
Create new dashboard
View your ingested data's key metrics, sample records, and ingestion history
Create dashboard
Close


Fabric
lms_practical_dataflowgen2
Lms_practical_lakehouse
LMS_KQL
LMS_KQL
Search
Trials activated: 51 days left
Eventhouse
Database
Table
Editing
Share
Explore data
Get data
Query with code
Real-Time Dashboard
KQL Queryset
Power BI report
Anomaly Detector
Notebook
View files
OneLake
Rename
Edit schema
Show ingestion mappings
Delete table
LMS_KQL
System overview
Databases
Monitoring
KQL databases
LMS_KQL_queryset
Tables
analytics-data
Ingestion Mappings
Update Policy
Timestamp
UserId
EventType
Status
Shortcuts
Materialized views
Functions
Database / Tables / analytics-data
Data Activity Tracker
Ingestion: 15 rows
Last run: 2:49:29 PM
1H 6H 1D 3D 7D 30D
Interval: 12 hours
May 19, 12:00 AM
May 19, 12:00 PM
May 20, 12:00 AM
May 20, 12:00 PM
May 21, 12:00 AM
May 21, 12:00 PM
May 22, 12:00 AM
Data preview
Schema insights
Search keyword
Timestamp | UserId | EventType | Status | IngestionTime
2025-01-01T09:05:00.000Z | U101 | Login | Success | 2026-05-22T09:18:56.3352929Z
2025-01-01T09:10:00.000Z | U102 | Login | Success | 2026-05-22T09:18:56.3352929Z
2025-01-01T10:00:00.000Z | U103 | Login | Failed | 2026-05-22T09:18:56.3352929Z
2025-01-01T10:15:00.000Z | U101 | FileUpload | Success | 2026-05-22T09:18:56.3352929Z
2025-01-01T10:30:00.000Z | U102 | FileDownload | Success | 2026-05-22T09:18:56.3352929Z
2025-01-02T09:00:00.000Z | U105 | Login | Success | 2026-05-22T09:18:56.3352929Z
2025-01-02T09:05:00.000Z | U101 | Login | Success | 2026-05-22T09:18:56.3352929Z
2025-01-02T10:20:00.000Z | U104 | FileUpload | Failed | 2026-05-22T09:18:56.3352929Z
2025-01-02T11:30:00.000Z | U106 | Login | Success | 2026-05-22T09:18:56.3352929Z
2025-01-03T09:15:00.000Z | U103 | Login | Success | 2026-05-22T09:18:56.3352929Z
2025-01-03T10:00:00.000Z | U106 | FileUpload | Success | 2026-05-22T09:18:56.3352929Z
2025-01-03T10:30:00.000Z | U101 | Logout | Success | 2026-05-22T09:18:56.3352929Z
2025-01-03T11:00:00.000Z | U104 | Login | Success | 2026-05-22T09:18:56.3352929Z
2025-01-03T11:20:00.000Z | U105 | Logout | Success | 2026-05-22T09:18:56.3352929Z
2025-01-03T12:00:00.000Z | U106 | Logout | Success | 2026-05-22T09:18:56.3352929Z
Top 15 records from the last 365 Days, limited by the hot cache retention
Table details
Compressed 2.1KB
Original 803B
OneLake
Availability Disabled
When enabled, all data in this table will be available in OneLake.
Learn more
Overview
Row count 15
Rows ingested last 24h 15
Schema last altered by fabric@tusharkolt...
Last ingestion May 22, 2026
Ingestion URI Copy URI
Caching policy Unlimited
Retention policy Unlimited
Related elements
Find by name
analytics-data_mapping (csv)


Fabric
lms_practical_dataflowgen2
Lms_practical_lakehouse
LMS_KQL
LMS_KQL
KustoQueryWorkbench_1
Saved
Search
Trials activated: 51 days left
Home
Help
Editing
Share
Save
Add data source
Copilot
Tab
Explorer
Search
LMS_KQL
Tables
analytics-data
EventType
Status
Timestamp
UserId
Materialized Views
Shortcuts
Entity Groups
Functions
Data sources
LMS_KQL
Run
Preview
Recall
Share query
Save to Dashboard
KQL Tools
Export to CSV
Power BI report
Add alert
["analytics-data"]
| count
Table 1
Add visual
Stats
Search
2026-05-22 09:36 (UTC)
Done (0.166 s)
1 records
Hide empty columns
Count
15

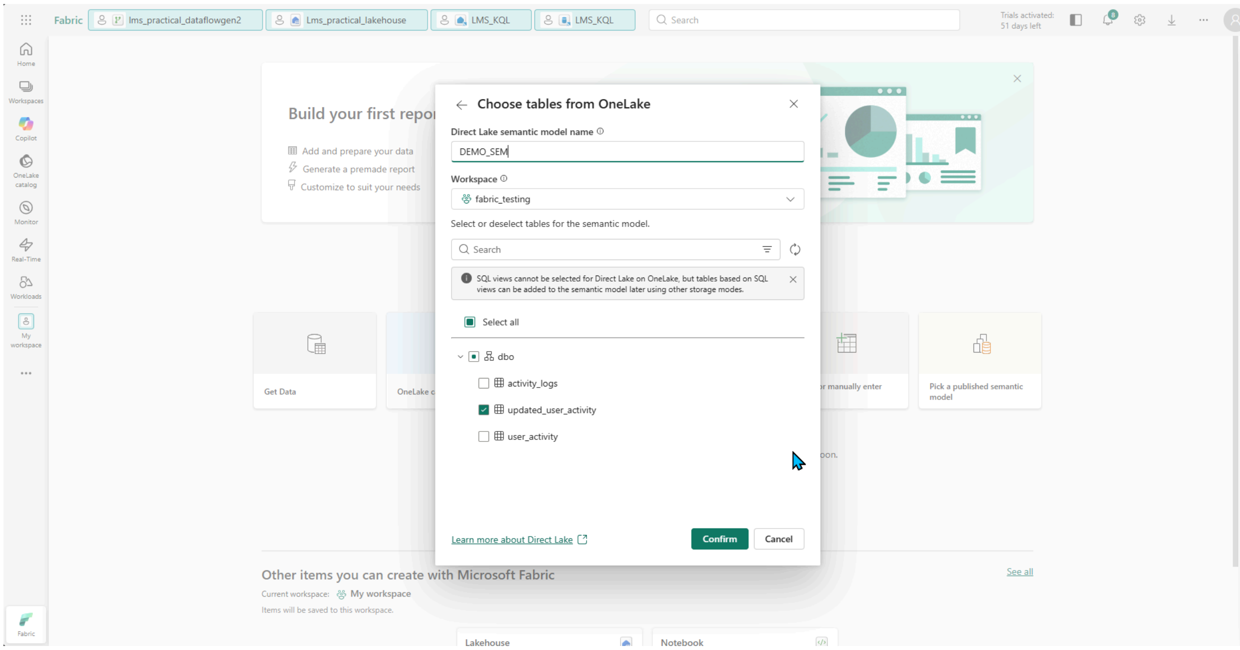


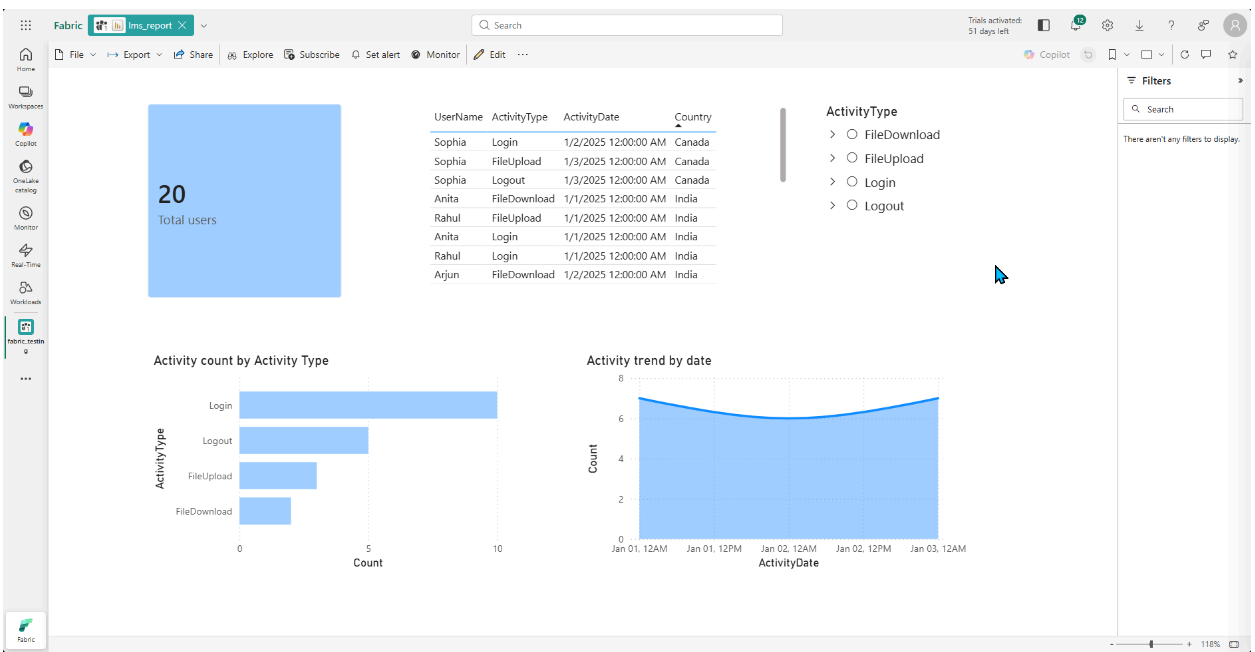


Task 4: Power BI Report (Fabric)

Create a one-page Power BI report inside Fabric using Dataflow Gen2 output with the following visuals:

- **Card:** Total Users
- **Bar Chart:** Activity count by Activity Type
- **Line Chart:** Activity trend by date
- **Table:** UserName, ActivityType, ActivityDate, Country
- **Slicer:** Country or ActivityType


Fabric
lms_practical_dataflowgen2
Lms_practical_lakehouse
LMS_KQL
LMS_KQL
Search
Build your first repor
Add and prepare your data
Generate a premade report
Customize to suit your needs
Choose tables from OneLake
Direct Lake semantic model name
DEMO_SEM
Workspace
fabric_testing
Select or deselect tables for the semantic model.
Search
SQL views cannot be selected for Direct Lake on OneLake, but tables based on SQL views can be added to the semantic model later using other storage modes.
Select all
dbo
activity_logs
updated_user_activity
user_activity
Learn more about Direct Lake
Confirm
Cancel
Get Data
Pick a published semantic model
Other items you can create with Microsoft Fabric
Current workspace: My workspace
Items will be saved to this workspace.
See all
Lakehouse
Notebook


Fabric
lms_report
Search
File
Export
Share
Explore
Subscribe
Set alert
Monitor
Edit
Copilot
Filters
There aren't any filters to display.
20
Total users
UserName ActivityType ActivityDate Country
Sophia Login 1/2/2025 12:00:00 AM Canada
Sophia FileUpload 1/3/2025 12:00:00 AM Canada
Sophia Logout 1/3/2025 12:00:00 AM Canada
Anita FileDownload 1/1/2025 12:00:00 AM India
Rahul FileUpload 1/1/2025 12:00:00 AM India
Anita Login 1/1/2025 12:00:00 AM India
Rahul Login 1/1/2025 12:00:00 AM India
Arjun FileDownload 1/2/2025 12:00:00 AM India
ActivityType
FileDownload
FileUpload
Login
Logout
Activity count by Activity Type
Login
Logout
FileUpload
FileDownload
ActivityType
Count
Activity trend by date
Count
ActivityDate
Jan 01, 12AM
Jan 01, 12PM
Jan 02, 12AM
Jan 02, 12PM
Jan 03, 12AM